

CS23334 Fundamentals of Data Science Lab

Experiment 15

Data Preprocessing Pipeline & Encoding in Data Science

Objective: To understand the end-to-end data preprocessing pipeline including missing value handling (mode, median, and mean imputation), categorical dummy encoding (`pd.get_dummies`), and binary target encoding.

Methodology:

- 1 **Initial Inspection & Missing Value Audit:** Load dataset (`pre-process_datasample.csv`) and inspect missing attributes across numerical and categorical features using `df.info()`.
- 2 **Multi-type Imputation Strategy:** Impute missing categorical attributes using feature mode (`Country.mode()[0]`), missing discrete attributes using feature median (`Age.median()`), and missing continuous attributes using feature mean (`Salary.mean()`).
- 3 **One-Hot Dummy Encoding:** Generate binary boolean indicators for categorical variables using `pd.get_dummies(df.Country)` and concatenate with numerical feature subsets using `pd.concat(..., axis=1)`.
- 4 **Target Encoding:** Transform binary target strings (No, Yes) into numeric integers (0, 1) using `.replace()`.

Code:

```
import numpy as np
import pandas as pd

df = pd.read_csv("pre-process_datasample.csv")
df

df.info()

df.Country.mode()
df.Country.mode()[0]
type(df.Country.mode())

df.Country.fillna(df.Country.mode()[0], inplace=True)
df.Age.fillna(df.Age.median(), inplace=True)
df.Salary.fillna(round(df.Salary.mean()), inplace=True)

df
```

```
pd.get_dummies(df.Country)
```

```
updated_dataset = pd.concat([pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]], axis=1)  
updated_dataset
```

```
df.info()
```

```
updated_dataset.Purchased.replace(['No', 'Yes'], [0, 1], inplace=True)  
updated_dataset
```

OUTPUT: 1. Initial Dataset, Mode Inspection & Imputed DataFrame

```
In [1]: df = pd.read_csv("pre-process_datasample.csv")
```

```
In [2]: df
```

```
Out[2]:
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

```
In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 10 entries, 0 to 9
```

```
Data columns (total 4 columns):
```

#	Column	Non-Null Count	Dtype
0	Country	9 non-null	object
1	Age	9 non-null	float64
2	Salary	9 non-null	float64
3	Purchased	10 non-null	object

```
In [4]: df.Country.mode()[0]
```

```
Out[4]: 'France'
```

```
In [5]: df.Country.fillna(df.Country.mode()[0], inplace=True)
```

```
In [6]: df.Age.fillna(df.Age.median(), inplace=True)
```

```
In [7]: df.Salary.fillna(round(df.Salary.mean()), inplace=True)
```

```
In [8]: df
```

```
Out[8]:
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	63778.0	Yes
5	France	35.0	58000.0	Yes
6	Spain	38.0	52000.0	No
7	France	48.0	79000.0	Yes
8	France	50.0	83000.0	No
9	France	37.0	67000.0	Yes

OUTPUT: 2. Dummy Feature Encoding & Final Target Encoded Dataset

```
In [9]: pd.get_dummies(df.Country)
```

```
Out[9]:
```

	France	Germany	Spain
0	True	False	False
1	False	False	True
2	False	True	False
3	False	False	True
4	False	True	False
5	True	False	False
6	False	False	True
7	True	False	False
8	True	False	False
9	True	False	False

```
In [10]: updated_dataset = pd.concat([pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]], axis=1)
```

```
In [11]: updated_dataset.Purchased.replace(['No', 'Yes'], [0, 1], inplace=True)
```

```
In [12]: updated_dataset
```

```
Out[12]:
```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	True	False	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1